

AIRPORT RUNWAY MANAGEMENT SIMULATOR

Week 3 Project Report
User Interface Development

Project	Airport Runway Management Simulator
Week	Week 3
Topic	User Interface Development
Technology	Java Swing

1. Introduction

The Airport Runway Management Simulator is a Java-based software project designed to simulate basic airport runway management activities. In Week 3, the project focuses on developing the graphical user interface (GUI). The main objective is to create a simple, organized, and user-friendly interface through which users can access different runway and flight management function.

2. Week 3 Objectives

The main objectives of Week 3 are:

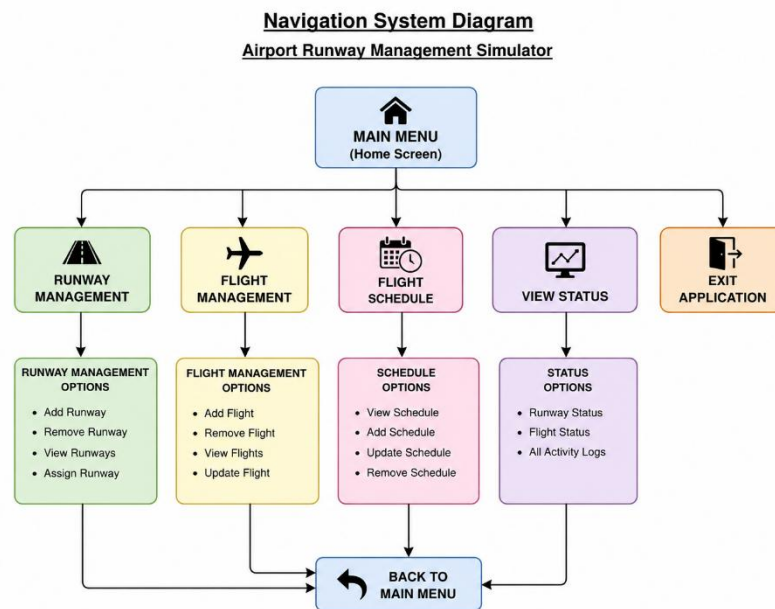
- Create the main application window using Java Swing.
- Design and implement the Main Menu.
- Add Runway Management options.
- Implement navigation between different screens.
- Handle button actions using Java event handling.

3. User Interface Development

The user interface provides the interaction layer between the user and the Airport Runway Management Simulator. Java Swing components are used to build the application window, menus, buttons, panels, and navigation controls.

4. Main Application Window

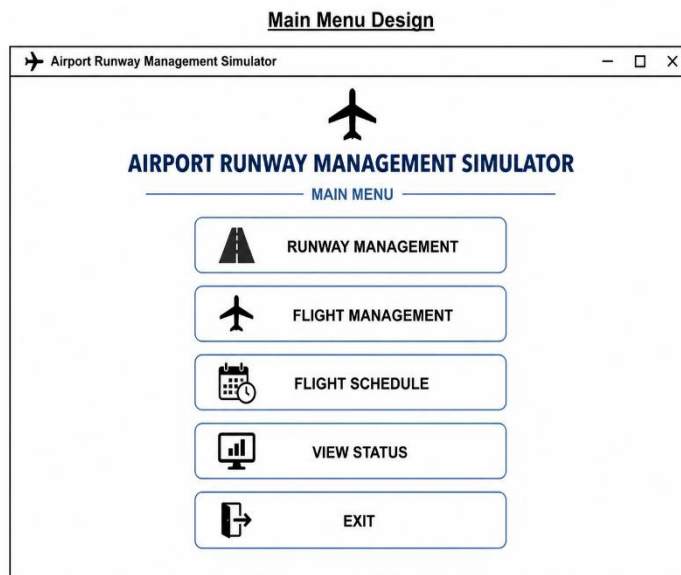
A JFrame is used as the main application window. It provides the basic structure of the GUI and acts as the container for the Main Menu and other screens. The window contains the project title and navigation buttons.



5. Main Menu Design

The Main Menu is the starting point of the application. It provides access to the major functions of the simulator. The planned menu options are:

- Runway Management
- Flight Management
- Flight Schedule
- View Status
- Exit



5. Runway Management Options

The Runway Management screen will provide options for managing airport runways. These options are designed to support the future simulator functions and provide a clear interface for runway-related operations.

Planned options include:

- Add Runway
- Remove Runway
- View Runways
- Assign Runway
- Back to Main Menu.



6. Navigation System

A navigation system is implemented through button event handling. When a user clicks a menu button,

the corresponding screen is opened. A Back button allows the user to return to the previous menu. This

creates a simple flow between the different GUI screens.

Example navigation flow: Main Menu → Runway Management → Runway Options → Back to Main Menu.

Navigation Flow Diagram
Airport Runway Management Simulator

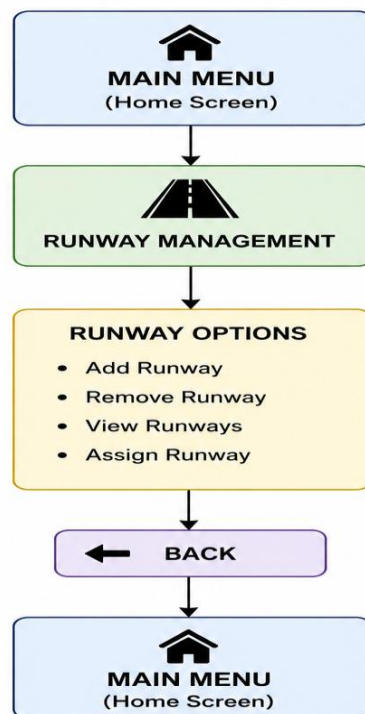


Figure 4: Navigation Flow of the Airport Runway Management Simulator

8. Technologies and Concepts Learned

Technology / Concept	Purpose
Java Swing	Used to develop the graphical user interface.
JFrame	Used to create the main application window.
JButton	Used to create clickable menu and navigation buttons.
Event Handling	Used to perform actions when buttons are clicked.
JPanel	Used to organize GUI components into separate sections.

9. GitHub Upload

The Week 3 implementation will be uploaded to the project GitHub repository. The required materials are:

- Main Menu Source Code
- GUI Design Files
- Related Java source files required to run the interface.

10. Demonstration

At the end of Week 3, the project should demonstrate a working Main Menu and a functional navigation system. The application should open successfully, display the Main Menu, respond to button clicks, open the Runway Management screen, and allow the user to return to the Main Menu.

11. Expected Outcome

After completing Week 3, the Airport Runway Management Simulator will have a basic working graphical interface. Users will be able to interact with the application through buttons and navigate between the main screens. This GUI will provide the foundation for implementing the simulator's functional features in later weeks.

12. Conclusion

Week 3 focuses on transforming the project structure into an interactive graphical application. By using Java Swing, JFrame, JButton, JPanel, and event handling, the project will provide a clear Main Menu and navigation system. The completed GUI will make the Airport Runway Management Simulator easier to use and ready for further feature development.